Asimov = median of toy pseudo-experiments

Checks for the histogram analysis

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The question

Does the no-fluctuation dataset actually tell us what a real experiment will see?

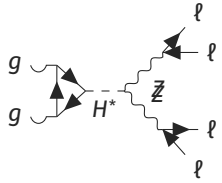
- To know how sensitive an analysis will be, we normally fit one fake dataset where nothing fluctuates — the Asimov dataset.
- That's supposed to stand in for running thousands of toy pseudo-experiments and taking the median. Much cheaper, if it's true.
- The catch: that substitution is a large-sample result. Off-shell Higgs at 1 fb^{-1} is about six events — not obviously a large sample.
- So before trusting it for anything fancier, we wanted to actually check it, carefully, for the simplest analysis we have: a plain histogram fit.

> This is a check, not a result talk. A lot of it is us thinking out loud about whether we're even testing the right thing.

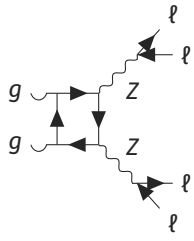
The process

Off-shell $gg \to H^* \to ZZ \to 4\ell$, at parton level.

We can get to 4ℓ two ways: through the Higgs, or not at all. Same final state, so they interfere.



signal



background

> Signal and background aren't separable event by event — the interference term is part of what any fit has to deal with, histogram or not.

The setup

One parameter, one histogram, lots of toys.

- Every event carries its matrix elements, so we can reweight it to any signal strength μ exactly — no detector simulation.
- Fit parameter: just μ. No nuisance parameters here.
- For a toy, we let every event weight fluctuate on its own (Poisson noise) and refit. 2000 toys per luminosity point, generated at $\mu_{\text{true}} = 1$.
- μ scanned over $[0, 4]$; luminosities from 1 to 3000 fb^{-1}.

Expected events at $\mu = 1$

	1 fb^{-1}	100 fb^{-1}	1000 fb^{-1}
signal	0.31	31.0	309.9
background	5.95	595.3	5953.0
interference	−0.61	−60.6	−606.4
total	5.66	565.7	5656.5

At 1 fb^{-1} the whole experiment is **six events** — the signal is a third of one of them.

Small numbers. Worth keeping in mind for everything that follows.

Ask 1: subtract each toy's own minimum first?

What we compare, before we compare it.

- Right now, before comparing a toy to Asimov, we shift the toy's whole curve down by its *own* best fit: we look at $t(\mu) = -2 \ln L(\mu) - \left(-2 \ln L(\hat{\mu}) \right)$, not the raw $-2 \ln L(\mu)$.
- The reasoning: each toy's curve sits at a random overall height that has nothing to do with μ — it's just how lucky that particular toy got. Subtracting the minimum removes that.
- Tae's original slides didn't do this — they took the median of the raw curves directly, offset and all.

Ask: Is subtracting each toy's own minimum the right move, or is there a reason to compare the raw curves instead?

Ask 2: should the curves match, or just the quantiles?

Even after that subtraction, we're not sure "Asimov = median" is the right target.

- Our rough back-of-envelope guess, if the usual large-sample assumptions hold: the *median* toy curve shouldn't sit exactly on the Asimov curve — it should sit a bit above it near μ_{true}, and be a bit wider, just from basic χ^2 statistics.
- If that's right, then maybe "Asimov = median of toys" was never quite the thing to check curve-by-curve. Maybe what should actually line up is the *spread* of toy curves at each μ — the 16%/50%/84% quantiles — against what Asimov predicts for that spread.
- We haven't fully convinced ourselves this is the right framing.

Ask: Should we be checking curve-on-curve agreement, or agreement of the quantiles at each μ? Is there a convention you'd expect us to use here?

Histogram results

PLACEHOLDER

Waits on the verification walkthrough and the reworked figures (`scripts/make_talk_figures.py`). Content: the measured toy spread against what Asimov predicts, at a few luminosities, plus an honest read of where it lines up and where it doesn't.

Backup